

Johan Sebastian Ospina Buitrago

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EXPERIENCE

Markforged

Oct 2022 – Present

R&D + Senior Machine Learning Engineer

Boston, MA

- Led cross-functional delivery of a key Computer Vision feature, aligning hardware-software roadmaps and engineering a projective geometry pipeline that scaled asynchronous image I/O to gigabyte levels with sub-1-minute end-user latency.
- Built and scaled an image-to-3D model search system using DINOv3 embeddings and a C++ Filament renderer to deliver sub-500ms query results; owned end-to-end model management, embedding generation, PostgreSQL pgvector storage, and search tooling.
- Maintained a terabyte-scale data ingestion pipeline and implemented data-driven material-tuning workflows, helping slash validation timelines from two quarters to one month.
- Streamlined internal analysis tools via cloud-based execution, eliminating local dependency bloat and enabling cross-functional team members to run evaluations on demand.
- Automated internal engineering tasks by developing agentic AI workflows (Context + Skill Creation), improving team execution speed and significantly reducing manual operational overhead.
- Productionized ML image classification models using Amazon SageMaker, MLflow, and Hugging Face, while architecting AWS Lambda pipelines for composite image generation via computational photography and OpenCV.
- Contributed to a patented 3D printing methodology in collaboration with cross-functional R&D partners, while mentoring junior engineers on core technical practices.

PTC - Vuforia

June 2021 – Sep 2022

Senior Prototyping Engineer

Boston, MA

- Minimized positional drift from SLAM-to-Context-Graph data structure in Unity and C# Mobile frameworks and AngularJS/Lit Element Web Frontend using Graph Isomorphisms
- Reduced technical debt in production codebases by optimizing build tools, including configuring linters, formatters, and CI/CD pipelines for improved code quality and development efficiency in a team of 5 engineers

Princeton University

Dec 2019 – May 2021

Research Assistant

Princeton, NJ

- Developed a pipeline using Amazon Web Services, Internal Princeton Infrastructure, and Database systems to gather quality image correspondence data from globally distributed users
- Engineered a [Python Software Pipeline](#) 🔗 to reproduce [research](#) 🔗 on calculating Camera Intrinsic Parameters using Image Correspondences and Projective Geometry

Wayfair

Internship 2016 + June 2017 – March 2019

R&D Software Engineer II

Boston, MA

- Led research into emerging immersive shopping technologies, exploring WebXR, ARKit, ARCore, Unity, BabylonJS, and automated material conversion from ray-traced assets to PBR workflows.
- Designed, implemented, and published [Doll House Projection Mapping Experience](#) 🔗 at CHI EA '20 with a team of 3 employees
- Wrote OpenGL Shading Language (GLSL) GPU Vertex and Fragment Shader Code for Real Time Graphics Applications

EDUCATION

Princeton University

Sep 2019 – May 2021

Master of Science in Computer Science

Princeton, NJ

Boston University

Sep 2013 – May 2017

Bachelor of Science in Computer Engineering, Magna Cum Laude

Boston, MA

SKILLS

Programming Languages : Python, TypeScript, JavaScript, C#, $\LaTeX$, Java, Bash, C++, Swift, SQL

Libraries/Frameworks : Node.js, React, ESLint, Jest, Vitest, Redux, Mocha/Chai, PyTorch, NumPy, Pandas, Parquet, OpenCV, Plotly, MLflow, Scikit-Learn, Jupyter, Matplotlib, CUDA, Zustand

Tools/Platforms : Git, GitHub, VS Code, Cursor, Unix, Unity, AWS Lambda, Docker, CircleCI, AWS Athena, AWS ECR, AWS Kinesis, AWS SageMaker, Hugging Face, ClickHouse, Kubernetes, PostgreSQL, Datadog, Amazon Bedrock, Esbuild, Redis, Terraform

Domain Knowledge : Optimization Techniques, Linear Algebra, 3D Printing, Machine Learning, Neural Networks, Computer Vision, Computer Graphics, Computational Photography, Edge Computing, Getting Buy-In, Deep Learning, NLP, Time Series Analysis, Model Deployment, A/B Testing, MLOps, Statistical Analysis, Data Engineering, Distributed Computing

Languages : Spanish, French